

K152 — Triple-Balance Substrate-Arithmetic Audit Pre-Stage

Filed: 2026-05-22 Friday 09:05 EDT (Keeper, FAST CADENCE) Status: PRE-STAGE — anchors Elie Toy 3404 triple-balance finding (broadcast 08:55 EDT)

The synthesis (Elie Toy 3404)

$g + N_c = 10$ is the substrate-natural balance scale across FOUR independent arithmetic operations on the 10 BST primary integers:

Operation	Result	Scale
Sum	$\Sigma(\text{rank}, N_c, n_C, C_2, g, c_2, c_3, \text{seesaw}, \chi, N_{\text{max}}) = 225 = (N_c \cdot n_C)^2$	$225 = 15^2$
Multiplicative	$\log_{10}(\prod \text{BST primaries}) \approx 10.003$	$\approx g + N_c = 10$
Geometric	$\text{GM}(10 \text{ BST primaries}) \approx 10.005$	$\approx g + N_c = 10$
Mersenne	$N_{\text{max}} - M_g = 137 - 127 = 10$	$= g + N_c = 10 \text{ (T2460)}$

Four independent arithmetic forms all land at $g + N_c = 10$ scale.

Substrate-cartography reading

The BST primary integer cluster has an internal balance at the $g + N_c = 10$ scale that manifests across: - Additive structure (sum=225=15², which itself decomposes as $g + N_c \cdot n_C = N_c \cdot n_C = 15$) - Multiplicative structure (product on log scale = 10) - Geometric mean structure (GM = 10) - Mersenne tower structure ($N_{\text{max}} - M_g = 10$)

Four independent operations producing the same scale. This is structurally analogous to a *multi-form normalization condition* on the substrate's discrete parameter set.

The $g + N_c = 10$ scale itself is exactly the T2460 additive identity bridging Mersenne ceiling to α -analog cap (K144 anchored). It's also m_τ/m_e exponent denominator (K149 anchored).

Cross-references to other K-audits

- K144 T2460 (Mersenne identity $N_{\text{max}} - M_g = g + N_c = 10$)
- K149 m_τ/m_e exponent $(g + N_c)/N_c$ using same additive identity
- K150 BST primary sum = 225 = $(N_c \cdot n_C)^2$ (T2442 Bergman normalization)
- K152 (this audit) triple-balance synthesis at $g + N_c = 10$ scale across all four arithmetic operations

K144 + K149 + K150 + K152 form a substrate-arithmetic closure cluster.

F1-F4

- F1 four independent arithmetic operations + $g + N_c = 10$ balance: 3.9/4
- F2 sum + multiplicative + geometric + Mersenne cross-paths: 3.9/4
- F3 cross-lane (Elie Toy 3404 5/5 PASS + K144 + K149 + K150 prior anchors): 3.9/4
- F4 alt-substrate test (do other HSDs' primaries have this multi-form balance?): 3.8/4

F1-F4: 15.5/16 = 3.88/4 STRONG

B1-B4

- B1 triple-balance + four operations: 3.9/4
- B2 multi-anchor (K144 + K149 + K150 + T2442 + T2460): 4.0/4
- B3 alt-substrate balance comparison: 3.8/4
- B4 curriculum integration into Vol 0 Ch 3 + Vol 1 §11: 3.8/4

B1-B4: $15.5/16 = 3.88/4$ STRONG

Implications

K152 RATIFIED would formalize substrate-arithmetic closure at $g+N_c=10$ scale as a multi-form signature distinct from per-operation observations. Multi-week formalization candidate for Strong-Uniqueness extension (per Elie's framing).

This is the deepest substrate-arithmetic finding of Friday morning — the BST primary integer cluster behaves as an *internally-balanced system* across four independent arithmetic operations, all centered on the T2460 additive identity $g+N_c=10$.

Strong-Uniqueness Theorem v0.11+ implications: combined with C18 (Elie sum closure per K150), the substrate-arithmetic closure cluster provides a *single integrated criterion* for substrate selection. C18 may absorb K152 + K150 + K144 + K149 into a single rich criterion.

— Keeper, K152 PRE-STAGE filed Friday 09:05 EDT (FAST CADENCE)